Presentation guide

Hybrid Movie Recommender · what to say on each slide

Sahil Lenka (2427010391) · **Kanishk Shukla** (2427010432)

Before you start

- **Open the live app ten minutes early.** Free hosting puts it to sleep when idle and it takes about a minute to wake. If you see a "Zzzz" page, click "Yes, get this app back up!".
- **Have slide 11 ready as the fallback.** It has screenshots of every screen, so the demo can be skipped entirely if the network fails.
- **Total length is about 13 minutes of speaking** plus questions. The demo on slide 10 is the part most likely to overrun, so keep it to the four actions listed there.
- **The speaker split below is a suggestion.** Whoever presents a slide should be the one who can answer questions on it, so swap if that fits you better.
- **Do not read these sentences word for word.** Read them a few times, then say them in your own words. The question-and-answer parts matter more than the script.

The three things to get across. One, the app works for a user with no history. Two, every pick carries a reason taken from the real score, not a generic sentence. Three, the user controls how adventurous the list is. If you only have two minutes, say those three and show the demo.

Who says what

You hand over every two slides, so each of you speaks six times and neither of you is quiet for long. Whoever presents a slide takes the questions on it. If a question lands on the other person's slides, pass it over rather than guessing.

Slides	Speaker	What it covers	Time
1 and 2	**Sahil**	Introduction, then the three problems and the goal	1.5 min
3 and 4	**Kanishk**	The six signals, then the data and how it is tested	2 min
5 and 6	**Sahil**	The pipeline, then the weights and the tuning	2 min
7 and 8	**Kanishk**	Results on full history, then cold start	2.5 min
9 and 10	**Sahil**	Accuracy against variety, then the live demo	2.5 min
11 and 12	**Kanishk**	Screenshots of the app, then limits and next steps	1.5 min
13	**Both**	Questions	the rest

That is roughly **6 minutes for Sahil** and **5.5 minutes for Kanishk**. Sahil opens and runs the demo; Kanishk owns the evidence, which means the results slides are his to defend.

Slide by slide

Slide 1. Title

Sahil · about 40 seconds

On screen: Project title, both names with registration numbers, course and university.

SAY

> *Good morning sir. I am Sahil Lenka and this is Kanishk Shukla. Our project is a hybrid movie recommender.*
>
> *It does three things that most recommenders do not. It works for a brand new user with no history, it tells you why each film was picked, and it lets the user decide how adventurous the list should be. It is a working web app and it is deployed, so we can open it live.*
>
> *Every number in this deck comes from an evaluation script in our repository, and we can run it in front of you.*

Slide 2. The problem

Sahil · about 50 seconds

On screen: Three problems with icons, and the goal in a box underneath.

SAY

> *Three problems drive the design.*
>
> *First, cold start. A new user has no ratings, so collaborative filtering has nothing to work with and everybody gets the same popular list.*
>
> *Second, no reasons. You get ten titles and no idea why, so you cannot tell a good recommendation from a lucky one and you stop trusting the system.*
>
> *Third, choosing takes too long, because catalogues are huge.*
>
> *So our goal was usable picks within a minute of arriving, each with a reason the user can check, and a control over how familiar or surprising the list is.*

IF SIR ASKS

Why is cold start actually hard?

Collaborative filtering learns from patterns of co-rating. With zero ratings there is nothing to correlate the user against, so the method has no signal at all.

Why not just show the most popular films to everyone?

That is one of our baselines. It is the same list for every user, so it ignores taste completely. We beat it by 47 percent on NDCG, which is on slide 7.

Slide 3. Six signals, one score

Kanishk · about 70 seconds

On screen: Six coloured cards, one per signal, and two lines explaining weighting and MMR.

SAY

> *We score every candidate film on six signals.*
>
> *Preference is the genres the user picked in the quiz. Content is TF-IDF over genre tokens and MovieLens tags, compared against the films they liked. Collaborative is item-item adjusted cosine on the ratings matrix. Context is tonight's mood mapped to a set of genres. Quality is a Bayesian average rating blended with how many people rated the film. Discovery rewards films outside the user's usual genres that still fit their taste.*
>
> *Each signal is min-max scaled inside the candidate set, so signals on different scales can be combined. Then they are weighted by how much we know about the user, and that is the next slide but one.*

IF SIR ASKS

Why scale each signal?

They have different natural ranges. Without scaling, whichever signal happens to produce the biggest raw numbers would dominate the ranking for no good reason.

Why a hybrid instead of one good method?

Each method fails somewhere. Content similarity needs films the user already liked. Collaborative filtering needs a lot of ratings and fails completely for a new user. Popularity is identical for everyone. Blending lets whichever signal has data carry the ranking at that moment.

What is adjusted cosine?

Cosine similarity computed after subtracting each user's mean rating, so a user who rates everything 4 or 5 does not distort the similarity. We also shrink it by n over n plus 10, so two films that share only a couple of raters count less.

Slide 4. Data and test setup

Kanishk · about 50 seconds

On screen: Four dataset numbers, then three rows describing the split, the hit definition and the baselines.

SAY

> *We used MovieLens ml-latest-small: 9,125 movies, 100,004 ratings, 671 users and 1,296 tag applications.*
>
> *One honest note. The specification mentions the 2018 release. That file was not reachable from our build machine, so we used the 2016 release. The format is identical and the code runs on either.*
>
> *For testing we split each user's ratings by time. The first 80 percent trains the models and the last 20 percent is held out. A held-out film rated 4 or higher counts as a hit. Everything, including item statistics and similarities, is built from training ratings only, so nothing from the test period leaks in.*
>
> *We compare against popularity only, content only, collaborative only, and our own hybrid using the untuned weights from the specification.*

IF SIR ASKS

Why split by time instead of randomly?

A random split lets the model see a user's later ratings while predicting their earlier ones, which leaks the future into the past. Splitting by time matches how the system would actually be used.

Why is a rating of 4 the cut-off for a hit?

On the 5 point MovieLens scale, 4 and above is a clear positive. Counting a 3 as a hit would reward recommending films the user was lukewarm about.

How many users are in each evaluation?

585 users have 20 or more training ratings and are used for the full-history test. 581 qualify for the cold-start test, which needs at least five highly rated films.

Slide 5. From signals to a ranked row

Sahil · about 60 seconds

On screen: Five numbered stages left to right, with the discovery dial settings underneath.

SAY

> *This is the pipeline for one row of recommendations.*
>
> *First we build the candidate set: every film with at least five ratings that the user has not already seen. Then hard filters remove disliked films, seen films and any avoided genre.*
>
> *Third, we score: the six signals times the stage weights, minus a small penalty for a film we have already shown several times without a click.*
>
> *Fourth, we re-rank with MMR, which I will explain on the results slide, and we cap any single genre at half the row.*
>
> *Fifth, the two largest signal contributions fill the reason template that the user sees on the card.*
>
> *The discovery dial changes two things at once: the MMR lambda, from 0.85 down to 0.55, and the discovery weight, from 0.05 up to 0.20.*

IF SIR ASKS

What is the repeat penalty?

If a film has been shown three or more times and the user has not clicked it, it loses 0.5 from its score so something else gets a turn.

Why cap a genre at half the row?

Without it, a strong genre preference fills all ten slots with one genre. The cap guarantees the row always shows at least two.

Slide 6. Weights by profile stage

Sahil · about 70 seconds

On screen: A table of the six signals against cold, warming and established, with the spec and tuned values side by side. Tuned values that changed are in red.

SAY

> *The weights depend on how much the system knows about the user. Cold means no interactions, warming is one to nineteen, established is twenty or more. Notice collaborative filtering is zero in the cold stage, because it genuinely has no signal there, and it grows as interactions arrive.*
>
> *This is the honest part of our method. The weights in the specification were untuned starting values, and with them our hybrid actually scored below plain popularity.*
>
> *So we tuned them, by random search on a validation slice cut out of the training data. The test split was never touched until the very end. The biggest change is a much larger collaborative weight.*
>
> *We also capped quality at 0.15. In our first search, without the cap, quality went to 0.40 and the app started giving a science fiction fan Shawshank and Pulp Fiction. That broke the specification's own rule that popularity is only a tiebreaker, so we capped it and tuned again.*

IF SIR ASKS

Is tuning the weights not cheating?

It would be if we tuned on the test set. We cut a validation slice out of the training data and tuned on that. The test split was used once, at the end, to produce the numbers on the next two slides.

Why is the cold column not tuned?

There are no interactions in the cold stage to tune against, so we left the specification values alone.

How did you search the weights?

Random search over weight vectors that sum to the available budget, with a floor of 0.05 for every signal so none can be switched off, scoring each candidate by NDCG at 10 on the validation slice.

Slide 7. Results: users with history

Kanishk · about 70 seconds

On screen: A bar chart of NDCG@10 for all seven methods, with three callouts on the right.

SAY

> *These are the results on the held-out last 20 percent, for 585 users.*
>
> *NDCG at 10 asks whether the films the user later liked appear near the top of the first ten. A liked film at rank one is worth more than the same film at rank ten, and the score is divided by the best possible ordering, so it runs from 0 to 1.*
>
> *The tuned hybrid beats popularity by about 47 percent, and the paired t-test across users gives p below 0.001, so that is a real difference and not noise.*
>
> *We want to be straight about two things. It does not beat collaborative filtering alone; the gap is about 1 percent and p is 0.63, so they are equal on this data. And with the untuned specification weights the hybrid scored 0.052, below popularity.*

IF SIR ASKS

If it only ties collaborative filtering, what is the point of the hybrid?

Collaborative filtering alone cannot serve a new user at all, it cannot explain a pick, and it gives the user no control. The hybrid matches it on accuracy while also doing those three things. The cold-start slide is where the hybrid clearly wins.

What does the paired t-test compare?

For each user we compute NDCG under both methods and test whether the mean difference across users is different from zero. Pairing by user removes the fact that some users are simply easier to predict than others.

Why is Recall@10 lower than you might expect?

Recall@10 divides by everything the user liked in the held-out period, which can be dozens of films. Ten slots cannot contain all of them, so the ceiling is low for active users. That is why we report it alongside NDCG rather than on its own.

Slide 8. Results: cold start

Kanishk · about 60 seconds

On screen: The same bar chart for the cold-start test, with three callouts.

SAY

> *This slide is the one that matters most for our original goal.*
>
> *We simulate a new user who has just finished the quiz. The model sees only their five best-rated films and the top three genres of those films. Nothing else from their history.*
>
> *In that setting the tuned hybrid is 27 percent better than popularity, with p equal to 0.002. Our specification target was 15 percent, so we are comfortably past it. It is also about ten times the content-only baseline, which has very little to work with from five films.*
>
> *If the user skips the quiz entirely, the weights collapse to the quality signal and they get popular, well rated films, which is a reasonable thing to show someone we know nothing about.*

IF SIR ASKS

Is five films a fair simulation of a new user?

It matches what our quiz actually collects: up to five favourites plus the genres. We also hide every film the user rated during training, for every method, so no method gets an unfair look at their history.

Does collaborative filtering work at all in this test?

Yes, but weakly. The five chosen films act as anchors for item-item similarity. That is why the warming stage still gives collaborative a weight instead of zero.

Slide 9. Accuracy against variety

Sahil · about 60 seconds

On screen: A table of NDCG, genre diversity and catalog coverage, with two callouts underneath.

SAY

> *Accuracy is not the only thing a recommender should be judged on, so we also measured variety.*
>
> *Diversity here is the average genre distance between the ten films in a list. MMR raises it by about 0.10, and it costs about 9 percent of NDCG. That is a real trade, not a free gain, and that is exactly why we made it a dial the user controls rather than a constant we chose for them.*
>
> *We also missed one target. Catalog coverage, the share of the catalogue the system ever recommends, is 13 percent against our 20 percent goal. The untuned weights reached 24 percent, but at much lower accuracy, so coverage and accuracy pull against each other here.*

IF SIR ASKS

Why keep MMR if it lowers accuracy?

Because an accurate row of ten near-identical films is a bad row to choose from. The accuracy loss is small and the user can turn it off by selecting Comfort zone.

How would you improve coverage?

The collaborative signal concentrates on well-rated popular films, so one option is a popularity penalty in the scoring, and another is the franchise cap we listed as future work. Both would cost some accuracy, which is the trade we just described.

Slide 10. Live example and demo

Sahil demonstrates, Kanishk narrates · about 90 seconds

On screen: A table of five picks with the reason shown for each, and the same user under Surprise me.

SAY

> *Here is a worked example. The user picked five science fiction favourites. On the left are the Comfort zone picks with the exact reason the app showed for each one.*
>
> *On the right, the same five favourites, nothing changed except the dial moved to Surprise me. Three of the five picks stay and two are swapped out for something further from the user's usual genres.*
>
> *Let me open the live app now.*

IF SIR ASKS

What to do in the demo

Skip the quiz. Show the four rows. Open Why this? on one card so the score breakdown appears. Press Not for me and point out the row re-ranks immediately. Move the dial to Surprise me and show the list change.

If the internet is down or the app is asleep

The app sleeps when idle and takes about a minute to wake, so open it before the presentation starts. If it will not load, slide 11 has screenshots of every screen, and the code runs offline with streamlit run app.py.

Slide 11. The app itself

Kanishk · about 30 seconds

On screen: Four screenshots of the running app with captions.

SAY

> *These are screenshots of the deployed app, not mockups.*
>
> *Top left is the quiz, and both questions are optional. Top right is the top pick with its reason and the score breakdown opened. Bottom left is a row of picks, each card carrying the reason it is there. Bottom right is the app showing its own offline evaluation, so the numbers we just presented can be checked inside the running system.*

IF SIR ASKS

Where do the posters come from?

From each film's English Wikipedia article, matched by exact article title, so no API key is needed. If a TMDB key is supplied the app uses TMDB instead, matched on the TMDB id in the dataset.

Slide 12. Limits and next steps

Kanishk · about 60 seconds

On screen: Two columns, limits on the left and next steps on the right.

SAY

> *What we are not claiming. Catalog coverage is 13 percent, below our own 20 percent target. On accuracy the hybrid ties collaborative filtering rather than beating it; the clear wins are over popularity and over content similarity. The dataset is small and older than the one in the specification, and the weights were tuned on a single split. And we have not tested with real users, so the online metrics in the specification are not measured.*
>
> *Next steps would be the full stack from the specification, training on MovieLens 25M, a franchise cap so a row cannot fill with sequels, and measuring pick rate with real users instead of a held-out split.*

IF SIR ASKS

Which part of the specification did you not build?

The Next.js and FastAPI stack, accounts, the franchise cap, group movie night, conversational search and availability data. We built the recommender, the evaluation and a working interface.

Slide 13. Questions

Both · about the rest

On screen: Closing slide with both names, registration numbers and the two links.

SAY

> *Thank you sir. The app is live at the address on screen and the full source, the tests and the evaluation script are on GitHub. We are happy to take questions.*

Harder questions

Did you write this code yourself?

Answer honestly, according to whatever your course requires. If AI assistance is permitted but must be disclosed, say so plainly and be ready to explain any part of the code, because that is what you will actually be assessed on.

Explain MMR in one sentence.

It picks each next film by its score minus its biggest similarity to the films already chosen, so the row covers more ground instead of repeating itself.

Why does NDCG use a logarithm?

The discount is 1 over log2 of the rank plus one. It makes a hit at rank 1 worth more than a hit at rank 10, but the difference between rank 9 and 10 is small, which matches how people actually scan a list.

What is the Bayesian average and why use it?

It pulls a film's mean rating towards the global mean in proportion to how few ratings it has. Without it, a film with one 5-star rating would outrank a film with 300 ratings averaging 4.5.

How do you know the explanation is honest?

The sentence is generated from the stored score breakdown, and a signal contributing less than 10 percent of the score is never named. We can open Why this? on any card and show that the numbers behind the sentence are the same numbers that produced the ranking.

What happens if a user dislikes everything?

Disliked films are removed from the candidate set, so it shrinks. If it ever empties, the app says so and points the user to the My taste page to undo some choices, instead of crashing.

How long does a recommendation take?

About 20 milliseconds for a row of ten on this dataset. The model is built once when the app starts and cached.

Why item-item collaborative filtering and not matrix factorisation?

Item-item is simple to explain, it works directly from the ratings we have, and it gives us the 'people who liked X also liked this' sentence for free. Matrix factorisation would be a reasonable next step, but its factors are harder to turn into a human-readable reason.

What would you do differently?

Tune on more than one split, measure coverage as an explicit objective rather than discovering it afterwards, and get the larger dataset in from the start.

Numbers worth memorising

Figure	Value	Where it comes from
Dataset	9,125 films, 100,004 ratings, 671 users	MovieLens ml-latest-small
Hybrid vs popularity, full history	+47% NDCG@10, $p < 0.001$	Slide 7
Hybrid vs popularity, cold start	+27% NDCG@10, $p = 0.002$	Slide 8
Hybrid vs collaborative only	+1%, $p = 0.63$, no real difference	Slide 7
Cost of MMR	9% of NDCG@10 for +0.10 diversity	Slide 9
Catalog coverage	13%, target was 20%	Slide 9
Evaluation users	585 full history, 581 cold start	Slide 4